

Total No. of Questions—8]

[Total No. of Printed Pages—2

Seat No.	
-------------	--

[5252]-174

S.E. (Information Technology) (I Semester) EXAMINATION, 2017
FUNDAMENTALS OF DATA STRUCTURES
(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

- N.B. :—**
- (i) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
 - (ii) Neat diagrams must be drawn wherever necessary.
 - (iii) Figures to the right indicate full marks.
 - (iv) Assume suitable data if necessary.

1. (a) What is structure in 'C' ? How do we declare the pointer of structure ? [3]
- (b) Explain with suitable example function call by reference and function call by value. [4]
- (c) Write a pseudo 'C' routine to multiply two matrices. [5]

Or

2. (a) What is a pointer variable ? Explain declaration, initialization and accessing a pointer variable with an example. [4]
- (b) Write and explain any *four* functions used for file handling. [4]
- (c) Write a pseudo C routine using pointer to add two matrices. [4]
3. (a) What is frequency count ? What is its importance in analysis of algorithm ? [4]
- (b) What is Abstract Data Type ? Explain ADT for an array. [4]

P.T.O.

- (c) What are Linear and Non-Linear Data Structures ? Explain with example. [4]
- Or*
4. (a) Sort the following numbers using insertion sort. Show all passes : [4]
60, 15, 80, 50, 40, 01, 07, 20
- (b) Write a pseudo code for binary search without recursion. [4]
- (c) Explain the time complexity for bubble sort method. [4]
5. (a) What is sparse matrix ? What are its applications ? [4]
- (b) Represent the following polynomials using arrays : [4]
- (i) $15x^3 - 7xy + y^2 - 50$
- (ii) $y^4 + 9y + 5$
- (c) Explain row major and column major representation of array. [5]
- Or*
6. (a) Write a pseudo C algorithm for addition of two sparse matrices. [5]
- (b) Define stack and queue with example. [4]
- (c) What is dynamic data structure ? List the advantages of Linked List. [4]
7. (a) Define and explain SLL, DLL & CLL with example. [6]
- (b) Write a pseudo code for the addition of a node after the position 'p' in singly linked list. [7]
- Or*
8. (a) Explain GLL with suitable example. [6]
- (b) Compare Array with Linked List. List the applications of Linked List. [7]